

ECD 415: Project Phase 1

Zeroth Presentation – July 2026

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Outline: Proposed Project Topics

- **Topic 1:** Edge AI Smart Battery Management System with Intelligent UPS
- **Topic 2:** Embedded Electronics Diagnostic Instrument
- **Topic 3:** Adaptive LoRa Mesh Emergency Communication Network

Presentation Structure per Topic

Introduction • Problem Statement • Proposed Methodology • Block Diagram
• Expected Outcomes • Applications

Topic 1

Edge AI Smart Battery Management System with Intelligent UPS

Topic 1: Introduction

- **Widespread Applications:** Lithium-ion battery packs are extensively used in electric vehicles (EVs), renewable energy storage, robotics, and industrial backup systems.
- **Limitations of Conventional BMS:** Provides only basic protection (over/under-voltage and over-current) with very limited insight into true battery health and long-term performance.
- **Intelligent BMS Upgrade:** Integrates Edge AI, predictive analytics, and smart UPS capability directly on an advanced ESP32-S3 platform.
- **Real-Time Intelligence:** Enables accurate SOC (State of Charge), SOH (State of Health), and RUL (Remaining Useful Life) estimation alongside pro-active on-device fault detection.

Topic 1: Problem Statement

- **Basic Protection Only:** Conventional BMS acts mainly as a simple hardware cutoff device without diagnostic intelligence or forecasting abilities.
- **Lack of Predictive Insights:** Unable to accurately evaluate cell degradation or predict Remaining Useful Life (RUL) under varying operating conditions.
- **Financial & Safety Risks:** Unprecedented cell aging leads to premature or delayed battery replacement, increasing overall lifecycle costs and sudden system failures.
- **Proposed Need:** An Edge AI-based Intelligent BMS for sophisticated predictive analytics, real-time telemetry, and autonomous UPS switching management.

Topic 1: Proposed Methodology

- **Data Acquisition:** Capture real-time cell voltage, current, and temperature using high-precision embedded sensors.
- **Battery Monitoring:** Implement rigorous parameter estimation and multi-layer hardware safety protection algorithms.
- **Edge AI Modeling:** Deploy lightweight machine learning models trained on historical cycling data for accurate SOH, RUL, and thermal fault prediction.
- **On-Device Inference:** Perform ML inference directly on an ESP32-S3 processor for deterministic, low-latency execution without relying on cloud servers.
- **Smart Management:** Facilitate intelligent UPS backup power switching, wireless telemetry data logging, and intuitive monitoring via a custom Flutter dashboard.

Topic 1: Block Diagram

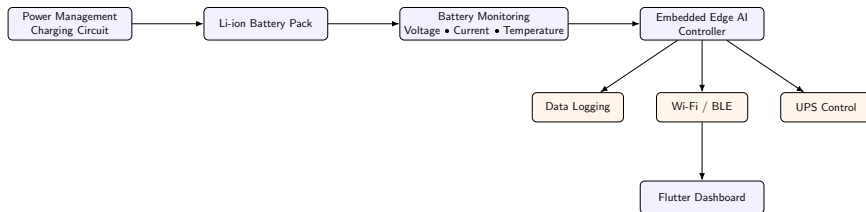


Figure 1: *Figure: Block diagram of the Edge AI Smart Battery Management System.*

Topic 1: Expected Outcomes

- A standalone, fully operational **Edge AI Smart Battery Management System**.
- High-precision **real-time monitoring** of critical cell voltage, load current, and pack temperature parameters.
- Accurate estimation algorithms for **State of Charge (SOC)** and **State of Health (SOH)**, alongside proactive **RUL forecasting**.
- Automated **on-device fault detection** linked to immediate, glitch-free UPS backup mode transitions.
- An intuitive, cross-platform **Flutter monitoring dashboard** displaying comprehensive analytical graphs and alerts.

Topic 1: Applications

- **Electric Vehicles (EVs):** Ensuring enhanced battery pack safety and precise driving range estimation.
- **Renewable Energy Storage:** Managing residential and municipal solar/wind battery banks efficiently.
- **Smart UPS Systems:** Uninterruptible power supplies with real-time health diagnostics and automated self-tests.
- **Industrial Robotics & AGVs:** Maximizing runtime and lifespan of autonomous factory fleet batteries.
- **Mission-Critical Backup:** Telecommunication base stations and medical ICU power backup units.

Topic 2

Embedded Electronics Diagnostic Instrument

Topic 2: Introduction

- **Multi-Functional Unit:** A compact, portable diagnostic instrument merging a digital oscilloscope, logic analyzer, and multi-protocol decoder into one standalone device.
- **Protocol Decoding Support:** Designed to analyze, inspect, and translate standard embedded communication protocols including UART, SPI, I2C, and automotive CAN buses.
- **Core Capabilities:** Facilitates real-time waveform visualization, precise automated signal parameter measurements, and sustained data logging.
- **Wireless Connectivity:** Transmits live high-speed diagnostic streams wirelessly to a companion smartphone/PC Flutter application for advanced analysis.

Topic 2: Problem Statement

- **High Cost & Bulkiness:** Commercial oscilloscopes and logic analyzers are highly expensive, heavy, and practically restricted to indoor lab work-benches.
- **Accessibility Gap:** Students, field technicians, and R&D engineers often struggle without portable, reliable debugging gear during field deployment or remote fieldwork.
- **Proposed Need:** An affordable, compact, handheld diagnostic instrument capable of capturing analog test waveforms and interpreting digital bus traffic simultaneously on a single embedded hardware platform.

Topic 2: Proposed Methodology

- **Signal Acquisition:** High-speed analog front-end circuitry captures and buffers analog and digital signals from the Device Under Test (DUT).
- **Embedded Processing:** An onboard microcontroller performs real-time sampling, hardware digital filtering, and structural protocol decoding.
- **Dual-Stage Visualization:** Basic signal indicators can be checked locally on the device, while raw stream data is broadcast over Wi-Fi / Bluetooth Low Energy (BLE).
- **Interactive Dashboard:** A custom Flutter software client receives the data stream, providing waveform zoom, trigger thresholds, hex protocol tables, and export capabilities.

Topic 2: Block Diagram

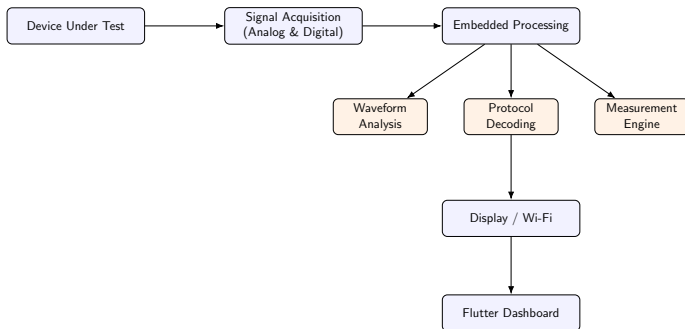


Figure 2: *Figure: Block diagram of the Embedded Electronics Diagnostic Instrument.*

Topic 2: Expected Outcomes

- A low-cost, pocket-sized **embedded diagnostic instrument** uniting digital oscilloscope and multi-channel logic analyzer functionalities.
- A dedicated **real-time protocol decoder** translating UART, SPI, I2C, and CAN messages into readable hexadecimal and text transcripts.
- Reliable **wireless transmission and data logging** directly to mobile devices or laptops via an aesthetically designed Flutter interface.
- An all-in-one educational and industrial troubleshooting tool ideal for hardware debugging and student lab instruction.

Topic 2: Applications

- **Embedded Firmware Development:** Verifying sensor timing, chip select signals, and SPI/I2C peripheral communication.
- **PCB Debugging & Maintenance:** Rapid diagnostic tracing on industrial control boards during field repairs.
- **Automotive Diagnostics:** Monitoring vehicular CAN bus networks, ECU traffic, and dashboard sensor telemetry.
- **Academic Laboratories & Robotics:** Empowering students to run oscilloscope experiments and test robotic circuit hardware anywhere.

Topic 3

Adaptive LoRa Mesh Emergency Communication Network

Topic 3: Introduction

- **Off-Grid Lifeline:** An Adaptive LoRa Mesh Network engineered to guarantee long-range emergency communication when conventional cellular towers and internet infrastructure collapse.
- **Autonomous Self-Organization:** Distributed handheld and stationary LoRa radio nodes autonomously discover peers and organize into a decentralised, self-healing wireless mesh topology.
- **Dynamic Packet Routing:** Messages and critical distress alerts dynamically hop across intermediary relay nodes to span large geographical disaster zones.
- **Integrated Features:** Embeds live GPS tracking, smartphone connectivity via Bluetooth, and dynamic priority routing for search-and-rescue teams.

Topic 3: Problem Statement

- **Infrastructure Vulnerability:** Conventional emergency response depends heavily on terrestrial cellular networks, fiber optic trunks, and grid power—all of which regularly fail during severe floods, earthquakes, and cyclones.
- **Isolation in Disaster Zones:** Affected communities and deployed first responders experience total communication blackouts, delaying critical search and medical rescues.
- **Proposed Need:** A fully portable, independent, self-healing, and energy-efficient long-range communication architecture capable of maintaining robust messaging without conventional telecom grids.

Topic 3: Proposed Methodology

- **Mesh Topology Discovery:** Each LoRa transceiver node continuously emits heartbeat broadcasts, exchanging routing metrics to establish a reliable, self-organizing mesh network.
- **Intelligent Multi-Hop Routing:** Distress broadcasts and chat texts are dynamically routed across optimal hop pathways toward active rescue command stations.
- **Smartphone Bluetooth Gateway:** Civilians and responders pair smartphones to local LoRa nodes via Bluetooth, composing text alerts through a native mobile application.
- **Central Command Telemetry:** A designated central command receiver consolidates incoming packets, displaying interactive GPS maps, SOS distress coordinates, and sensor alerts on a Flutter dashboard.

Topic 3: Block Diagram

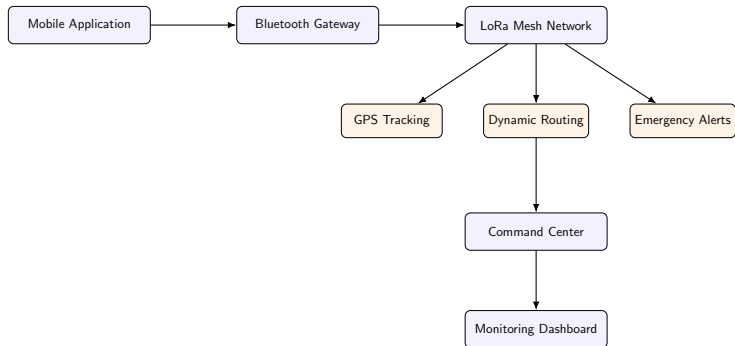


Figure 3: Figure: Block diagram of the Adaptive LoRa Mesh Emergency Communication Network.

Topic 3: Expected Outcomes

- A decentralised, functional **LoRa mesh communication network** governed by intelligent self-healing routing algorithms.
- Reliable **long-range, off-grid text messaging** and emergency broadcast capability without SIM cards or cellular connectivity.
- Real-time **GPS location tracking** and hazard sensor data aggregation across active mobile nodes in the disaster zone.
- An interactive **Flutter mobile client and command center GIS dashboard** for streamlined tactical disaster response.

Topic 3: Applications

- **Disaster Management & Rescue:** Lifeline communication during severe floods, earthquakes, landslides, and extreme typhoons.
- **Remote & Forest Surveillance:** Coordinating wild-land firefighters and tracking personnel across expansive forests without cellular coverage.
- **Tactical & Military Operations:** Establishing rapid, secure, ad-hoc wireless communication grids for reconnaissance deployments.
- **Smart Agriculture & Deep IoT:** Connecting remote agro-meteorological stations across expansive rural farmlands.

THANK YOU

Questions & Discussion